

①

1 Principal Moments of Inertia (1000, 1400, 1700) kg-m²

$$h = 500 \text{ km} \rightarrow \omega_0 = \sqrt{\mu / (R_E + h)^3} = 1.087 \times 10^{-3} \text{ rad/sec}$$

a) Based on the stability result in class,

$$I_x = 1.400 \text{ kg-m}^2, I_y = 1.700 \text{ kg-m}^2, I_z = 1.000 \text{ kg-m}^2$$

Such that $I_y > I_x > I_z$

b. Linearized EoM for pitch, roll & yaw in state-space form:

$$\begin{bmatrix} \dot{\Delta \theta} \\ \Delta \omega_y \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ \frac{3\omega_0^2 (I_z - I_x)}{I_y} & 0 \end{bmatrix} \begin{bmatrix} \Delta \theta \\ \Delta \omega_y \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{I_y} \end{bmatrix} M_y$$

$$\begin{bmatrix} \Delta \dot{\phi} \\ \Delta \dot{\omega}_x \\ \Delta \dot{\psi} \\ \Delta \dot{\omega}_z \end{bmatrix} = \begin{bmatrix} 0 & 1 & -\omega_0 & 0 \\ \frac{3\omega_0^2 (I_z - I_y)}{I_x} & 0 & 0 & \omega_0 \frac{I_y - I_z}{I_x} \\ \omega_0 & 0 & 0 & 1 \\ 0 & \omega_0 \frac{I_x - I_y}{I_z} & 0 & 0 \end{bmatrix} \begin{bmatrix} \Delta \phi \\ \Delta \omega_x \\ \Delta \psi \\ \Delta \omega_z \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ \frac{1}{I_x} & 0 \\ 0 & 0 \\ 0 & \frac{1}{I_z} \end{bmatrix} \begin{bmatrix} M_x \\ M_z \end{bmatrix}$$

E-values for pitch dynamics, $\lambda = \pm 0.9315 \times 10^{-3} j$

E-values for roll & yaw dynamics $\lambda = \pm 0.0017 j, \pm 0.0006 j$

which can be also determined from the char. eqn.

$$s^4 + (1 + 3k_1 + k_1 k_3) \omega_0^2 s^2 + 4k_1 k_3 \omega_0^4 = 0$$

where $k_1 = (I_y - I_z) / I_x, k_3 = (I_y - I_x) / I_z$

c.
$$\begin{bmatrix} \psi \\ \theta \\ \phi \end{bmatrix}_{t=0} = \begin{bmatrix} 0.1 \\ 0.1 \\ 0.1 \end{bmatrix} \text{ rad}, \quad \begin{bmatrix} \dot{\psi} \\ \dot{\theta} \\ \dot{\phi} \end{bmatrix}_{t=0} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \text{ rad/sec}$$

From the rotational kinematics at $t=0$,

$$\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} \dot{\psi} \\ \dot{\theta} \\ \dot{\phi} \end{bmatrix}_{t=0} = \begin{bmatrix} 0 & \sin\phi \sec\theta & \cos\phi \sec\theta \\ 0 & \cos\phi & -\sin\phi \\ 1 & \sin\phi \tan\theta & \cos\phi \tan\theta \end{bmatrix} \begin{bmatrix} \omega_x^r \\ \omega_y^r \\ \omega_z^r \end{bmatrix}$$

Since the rank of the matrix is 3 except $\theta = \frac{\pi}{2}$,

$$\begin{bmatrix} \omega_x^r \\ \omega_y^r \\ \omega_z^r \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Then, from the relative angular velocity dynamics, relations

$$\begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix}_{t=0} = \begin{bmatrix} \omega_x^r \\ \omega_y^r \\ \omega_z^r \end{bmatrix}_{t=0} + \omega_0 \begin{bmatrix} \cos\theta \sin\phi \\ \cos\phi \cos\psi + \sin\phi \sin\theta \sin\psi \\ -\sin\phi \cos\psi + \cos\phi \sin\theta \sin\psi \end{bmatrix}_{t=0}$$

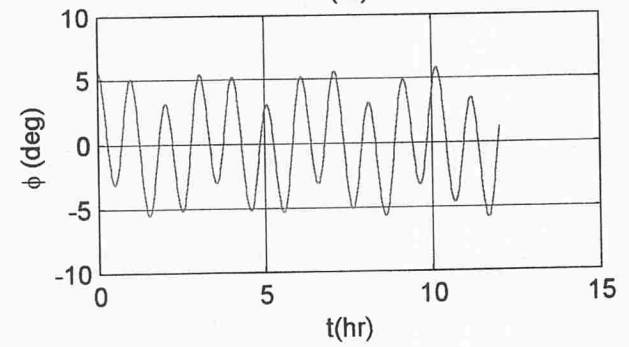
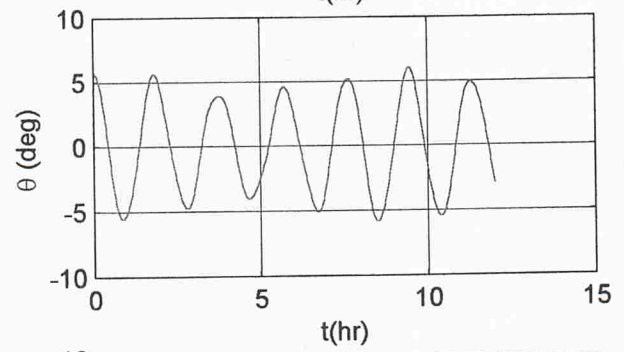
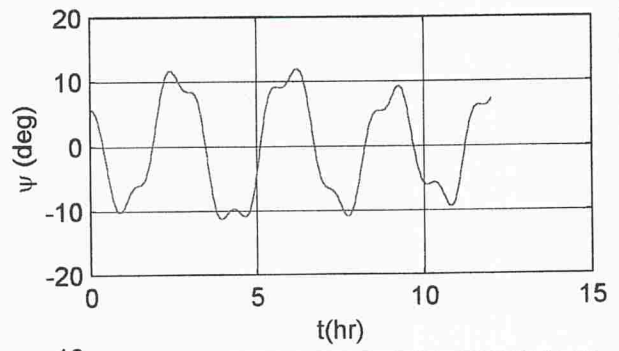
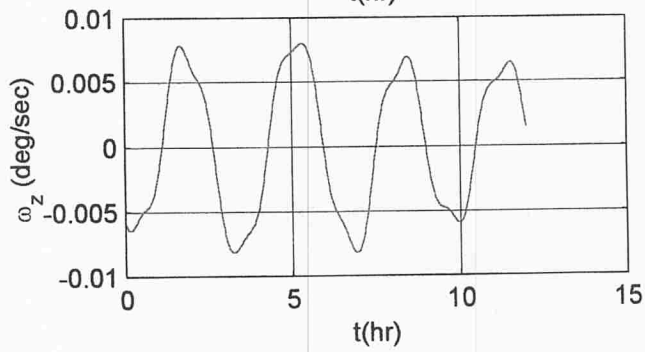
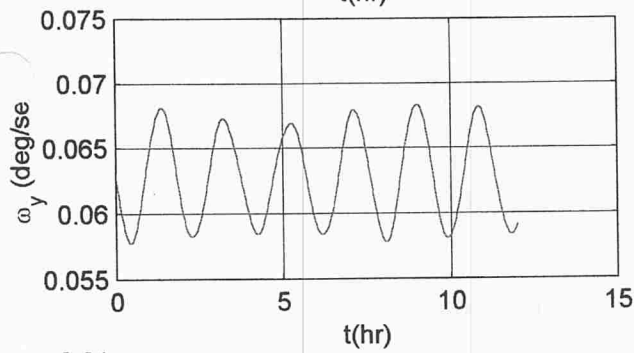
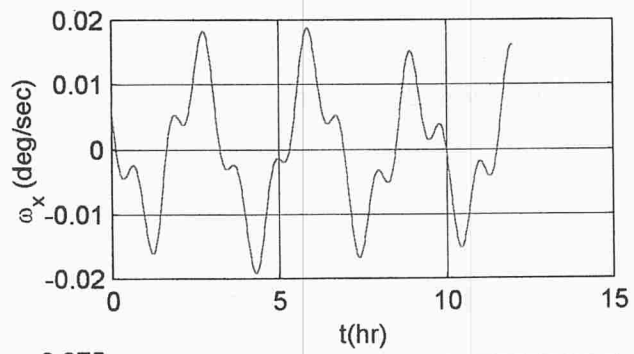
$$\Rightarrow \begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix}_{t=0} = \begin{bmatrix} 1.1014 \times 10^{-4} \\ 1.0988 \times 10^{-3} \\ -9.9142 \times 10^{-5} \end{bmatrix} \text{ rad/sec}$$

As is seen, the initial Euler Angle and initial angular velocity are very close to equilibrium point.

$$\begin{bmatrix} \psi \\ \theta \\ \phi \end{bmatrix}_{eq} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix}_{eq} = \begin{bmatrix} 0 \\ \omega_0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1.1087 \times 10^{-3} \\ 0 \end{bmatrix} \text{ rad/sec}$$

The following plots show that the behavior of nonlinear dynamics are similar to oscillatory behavior of linearized dynamics (attach plots)

3



2. $(I_x, I_y, I_z) = (1200, 1000, 800) \text{ kg-m}^2$

Since the S/C is placed into a geosynchronous orbit,

$$\begin{cases} \omega_0 = \frac{2\pi}{24 \times 60 \times 60 \text{ sec}} = 7.2722 \times 10^{-5} \text{ rad/sec} \\ h = (\mu/\omega_0^2)^{1/3} - R_E = 3.5871 \times 10^4 \text{ km} \end{cases}$$

a. Based on the given data.

$$\begin{bmatrix} \dot{\Delta \theta} \\ \Delta \dot{\omega}_y \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -6.3462 \times 10^{-9} & 0 \end{bmatrix} \begin{bmatrix} \Delta \theta \\ \Delta \omega_y \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \times 10^{-3} \end{bmatrix} M_y$$

$$\lambda = \pm 7.9663 \times 10^{-5} j$$

b.
$$\begin{bmatrix} \dot{\Delta \phi} \\ \Delta \dot{\omega}_x \\ \Delta \dot{\psi} \\ \Delta \dot{\omega}_z \end{bmatrix} = \begin{bmatrix} 0 & 1 & -7.2722 \times 10^{-5} & 0 \\ -2.6422 \times 10^{-9} & 0 & 0 & 1.2120 \times 10^{-5} \\ 7.2722 \times 10^{-5} & 0 & 0 & 1 \\ 0 & 1.8181 \times 10^{-5} & 0 & 0 \end{bmatrix} \begin{bmatrix} \Delta \phi \\ \Delta \omega_x \\ \Delta \psi \\ \Delta \omega_z \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 8.3333 \times 10^{-4} & 0 \\ 0 & 0 \\ 0 & 1.25 \times 10^{-3} \end{bmatrix} \begin{bmatrix} M_x \\ M_z \end{bmatrix}$$

$$\lambda = \begin{cases} 2.0329 \times 10^{-20} \pm 9.0971 \times 10^{\pm} j \\ \pm 2.3733 \times 10^{-5} \end{cases}$$

c.
$$\begin{bmatrix} \psi \\ \theta \\ \phi \end{bmatrix}_{t=0} = \begin{bmatrix} 0 \\ 0.1 \\ 0.1 \end{bmatrix} \quad \begin{bmatrix} \dot{\psi} \\ \dot{\theta} \\ \dot{\phi} \end{bmatrix}_{t=0} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Similarly as in Prob 1

$$\begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \text{ rad/sec} \Rightarrow \begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix}_{t=0} = \begin{bmatrix} 0 \\ 7.2359 \times 10^{-5} \\ -7.2601 \times 10^{-6} \end{bmatrix} \text{ rad/sec}$$

As is seen in b, the equilibrium for the linearized roll/yaw motion is unstable.

However, the plots on the next page show Not divergent But oscillatory behavior for the initial conditions given.

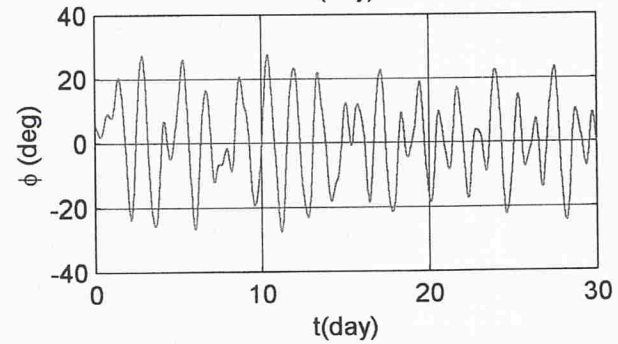
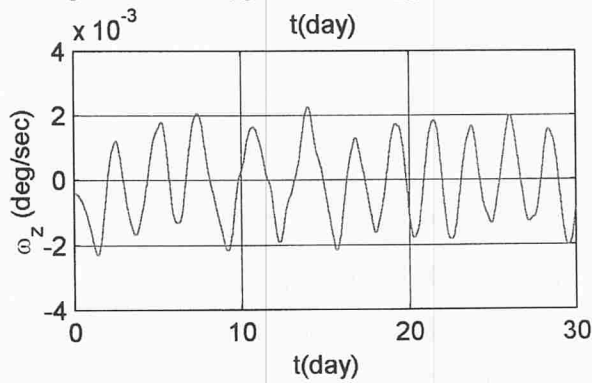
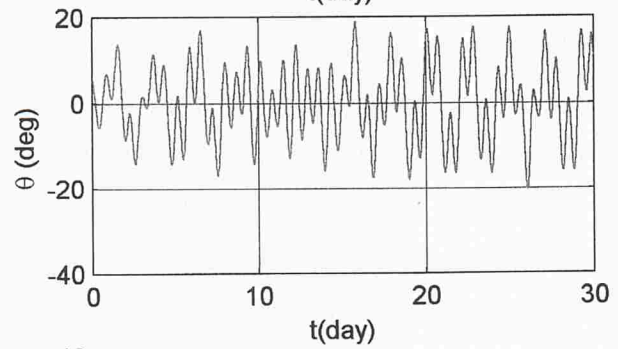
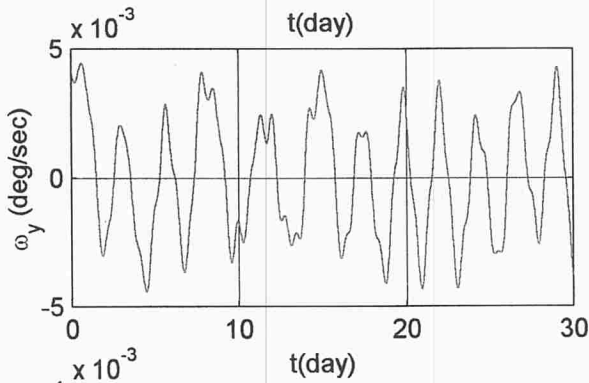
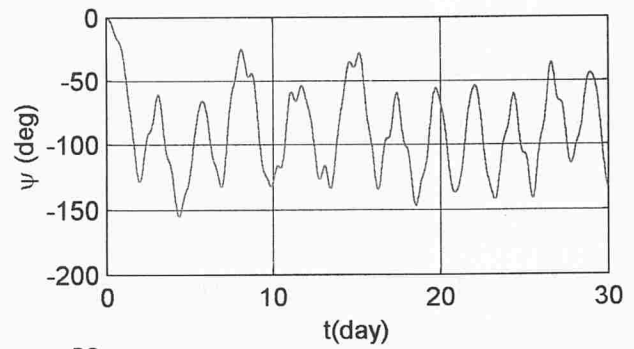
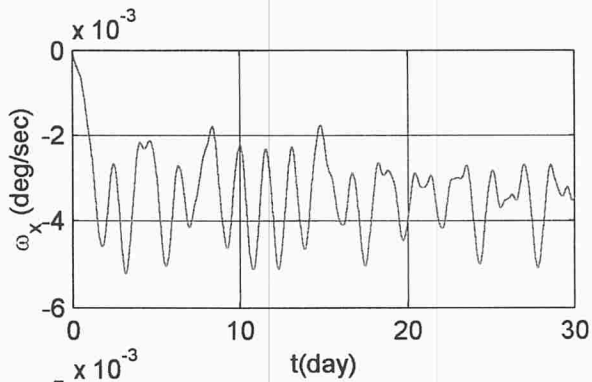
This should not be interpreted that the nonlinear dynamics are stable, but that the given initial condition happens to be located such that the trajectory does not diverge.

For example, if the initial condition is slightly different from the given one such that

$$\begin{bmatrix} \psi \\ \theta \\ \phi \end{bmatrix}_{t=0} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} \dot{w}_x \\ \dot{w}_y \\ \dot{w}_z \end{bmatrix} = \begin{bmatrix} 7.2359 \times 10^{-5} \\ 7.2359 \times 10^{-5} \\ -7.2601 \times 10^{-6} \end{bmatrix} \text{ rad/sec}$$

the roll/yaw motion shows divergent behavior
(attach plots)

$$\begin{bmatrix} \psi \\ \theta \\ \phi \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix} = \begin{bmatrix} 0 \\ 7.235 \times 10^{-5} \\ -7.260 \times 10^{-6} \end{bmatrix}, \text{ which is given at the problem}$$



7

$$\begin{bmatrix} \psi \\ \theta \\ \phi \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix} = \begin{bmatrix} 7.235 \times 10^5 \\ 7.235 \times 10^5 \\ -7.260 \times 10^6 \end{bmatrix}$$

which is slightly different from the given initial condition at the problem

ok

